

Physical Chemistry (I)

Lecture 20

Solid-Liquid Equilibria

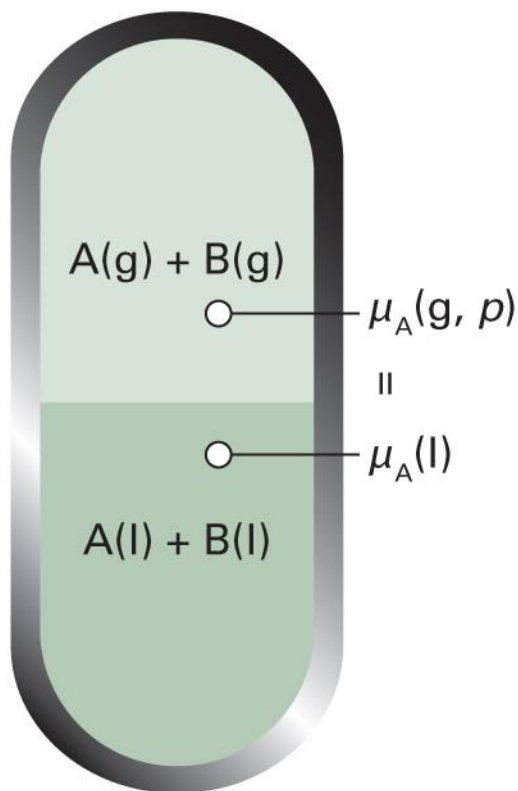


In the Last Class

- Phase diagrams of ideal solutions
 - Compositions of liquid and gas phases
 - Composition-pressure diagrams
 - Composition-temperature diagrams
- Phase diagrams of real solutions
 - Azeotropes
 - Phase separation



Liquid-Vapor Equilibrium



- Constraints

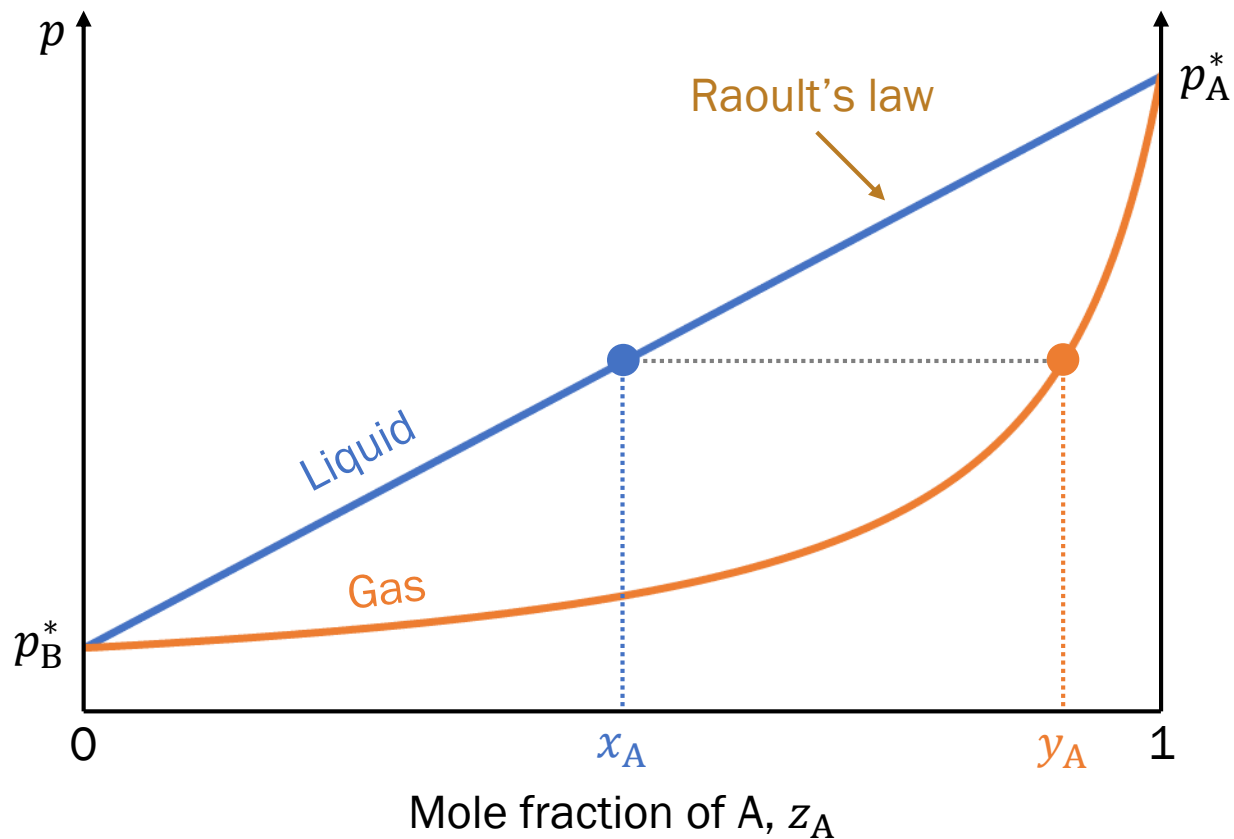
$$\mu_A(g) = \mu_A(l)$$

$$\mu_B(g) = \mu_B(l)$$

- May have $x_A(g) \neq x_A(l)$

Last class

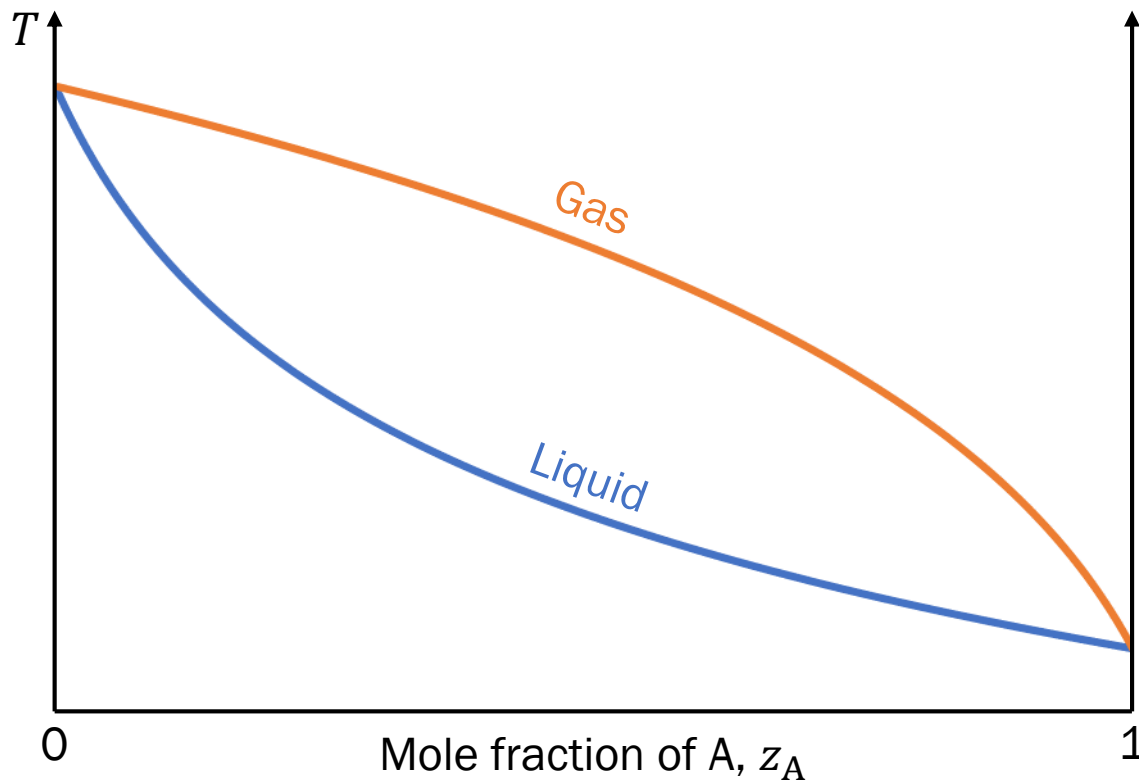
Composition-Pressure



Last class

Composition-Temperature

$$p_A^* > p_B^* \text{ (A is more volatile)}$$



Real Solutions

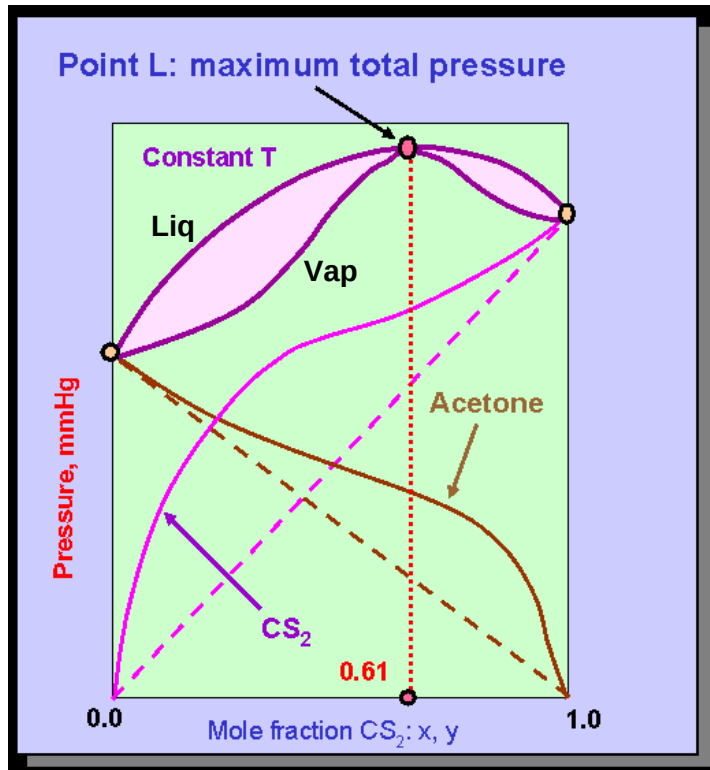
$$\xi = \frac{z}{k_B T} \left(w_{AB} - \frac{w_{AA} + w_{BB}}{2} \right) \neq 0 \text{ (BW model)}$$

- $\xi > 0$: AB interactions are weaker
 - $\xi < 0$: AB interactions are stronger
1. Change in liquid-gas equilibrium: azeotrope
 2. Change in liquid-liquid equilibrium: phase separation

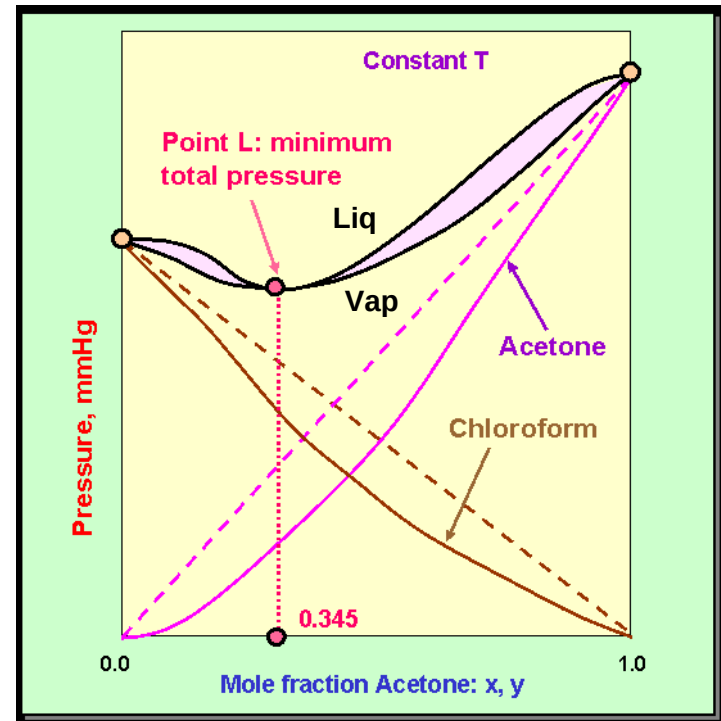
Last class

Composition-Pressure

$$\xi > 0$$



$$\xi < 0$$

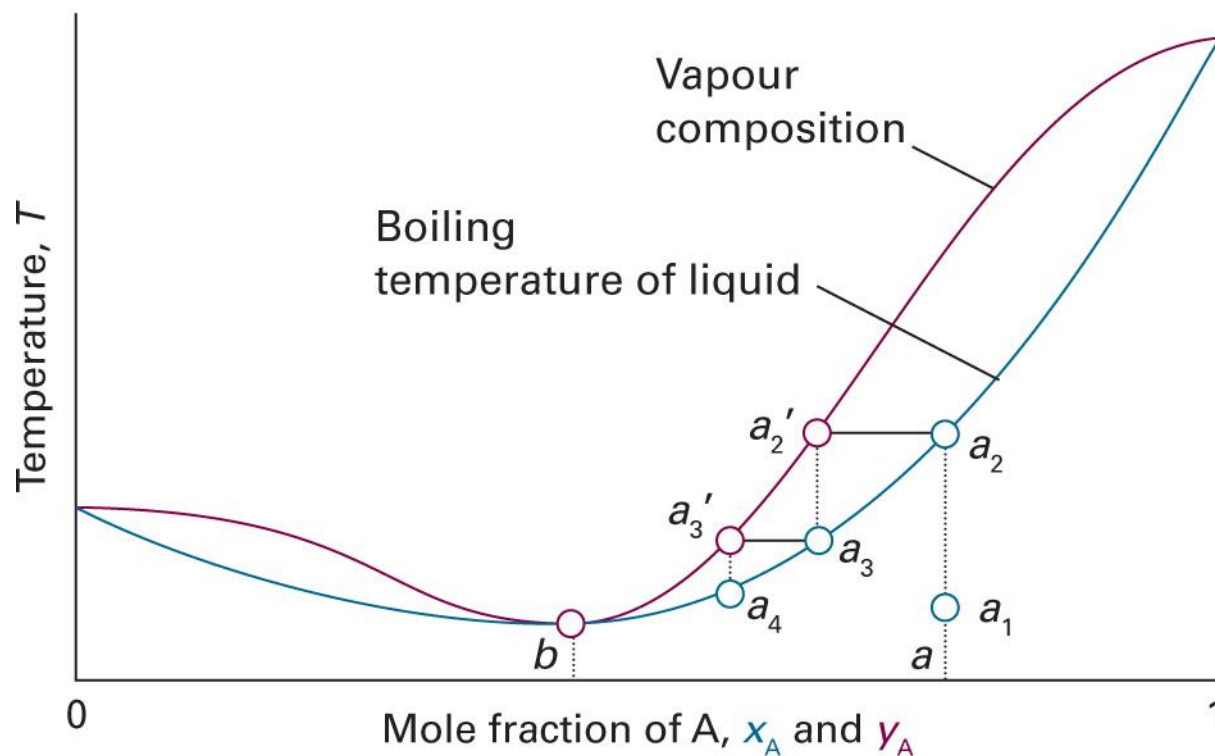


(Image: http://www.separationprocesses.com/Distillation/DT_Chp01f.htm)

Last class

Low-Boiling Azeotrope

$$(\xi > 0)$$

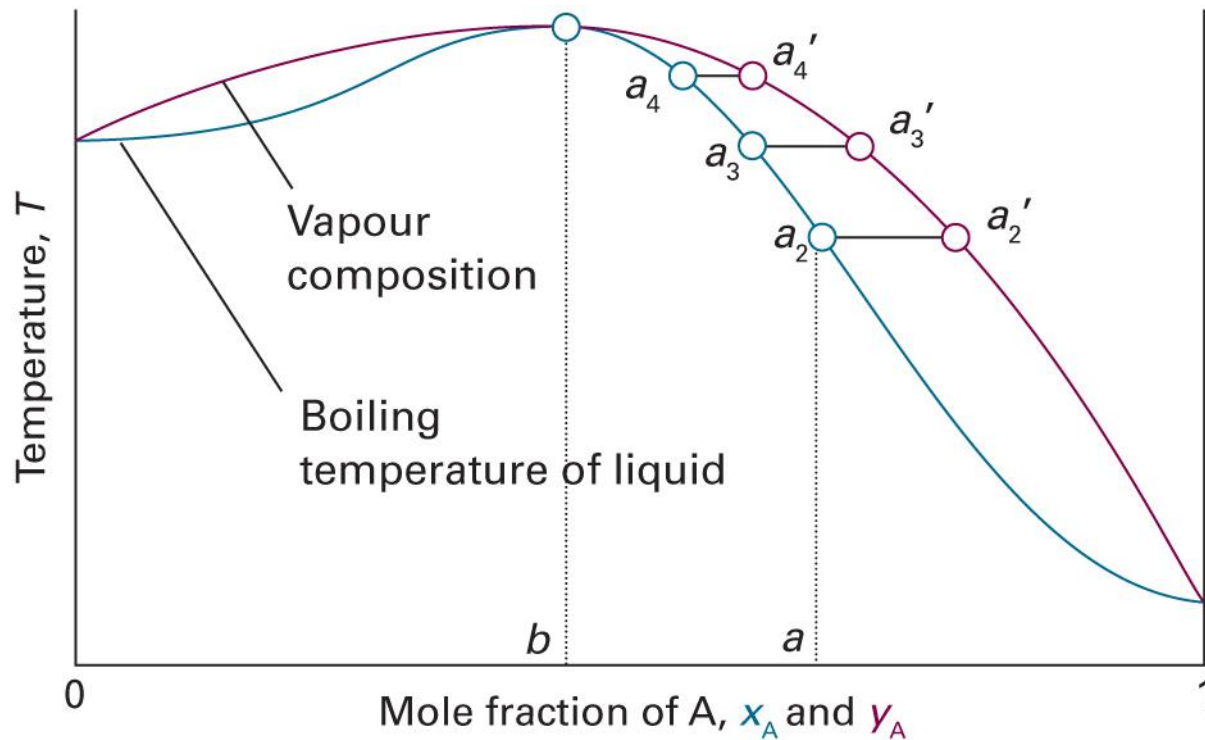


(Image: Atkins *et al.*, *Atkins' Physical Chemistry*, 12th ed., Figure 5C.12)

Last class

High-Boiling Azeotrope

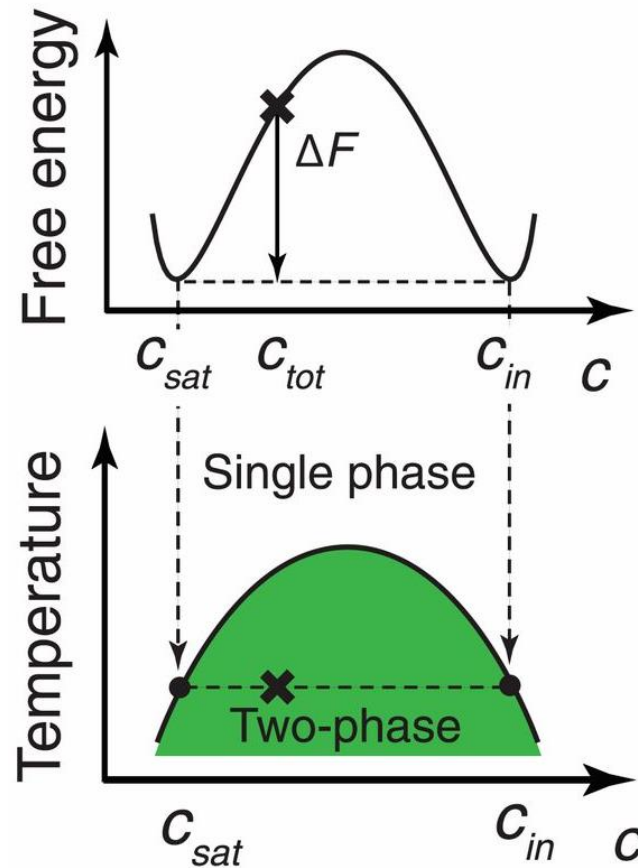
$$(\xi < 0)$$



(Image: Atkins *et al.*, *Atkins' Physical Chemistry*, 12th ed., Figure 5C.11)

Last class

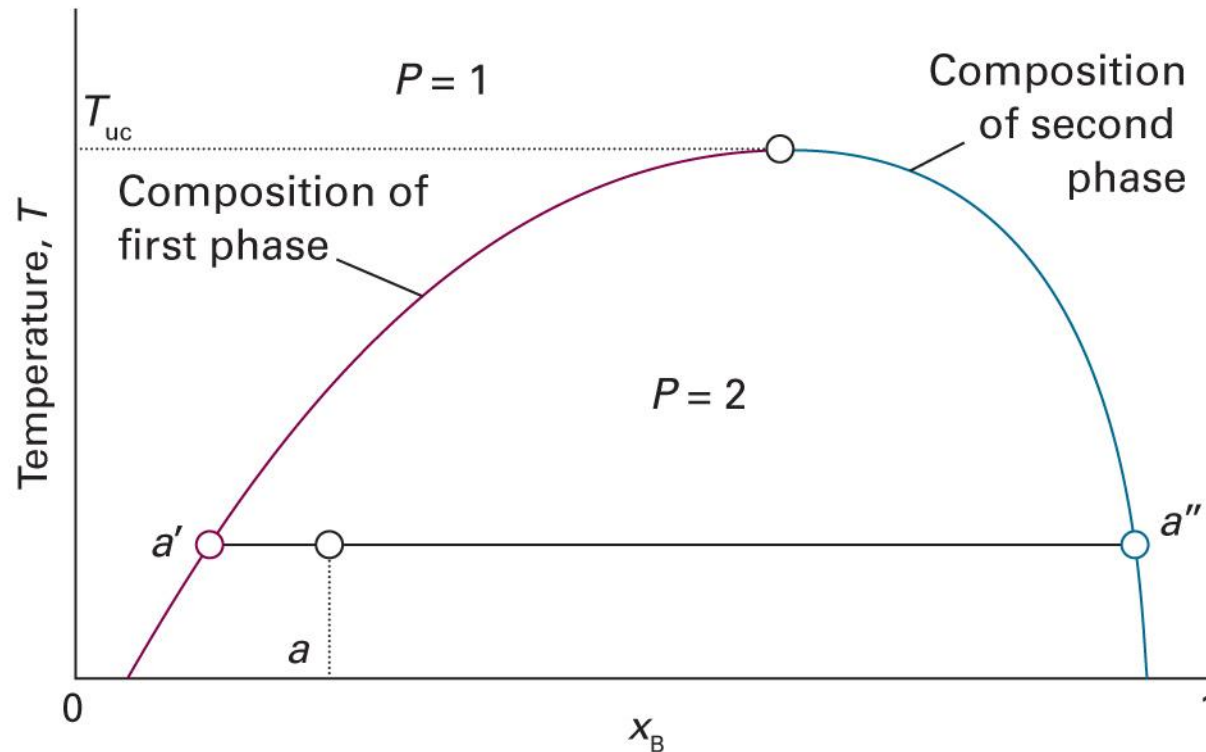
Phase Separation



(Image: Shin and Brangwynne, *Science* 357: eaaf4382 (2017), Figure 3)

Last class

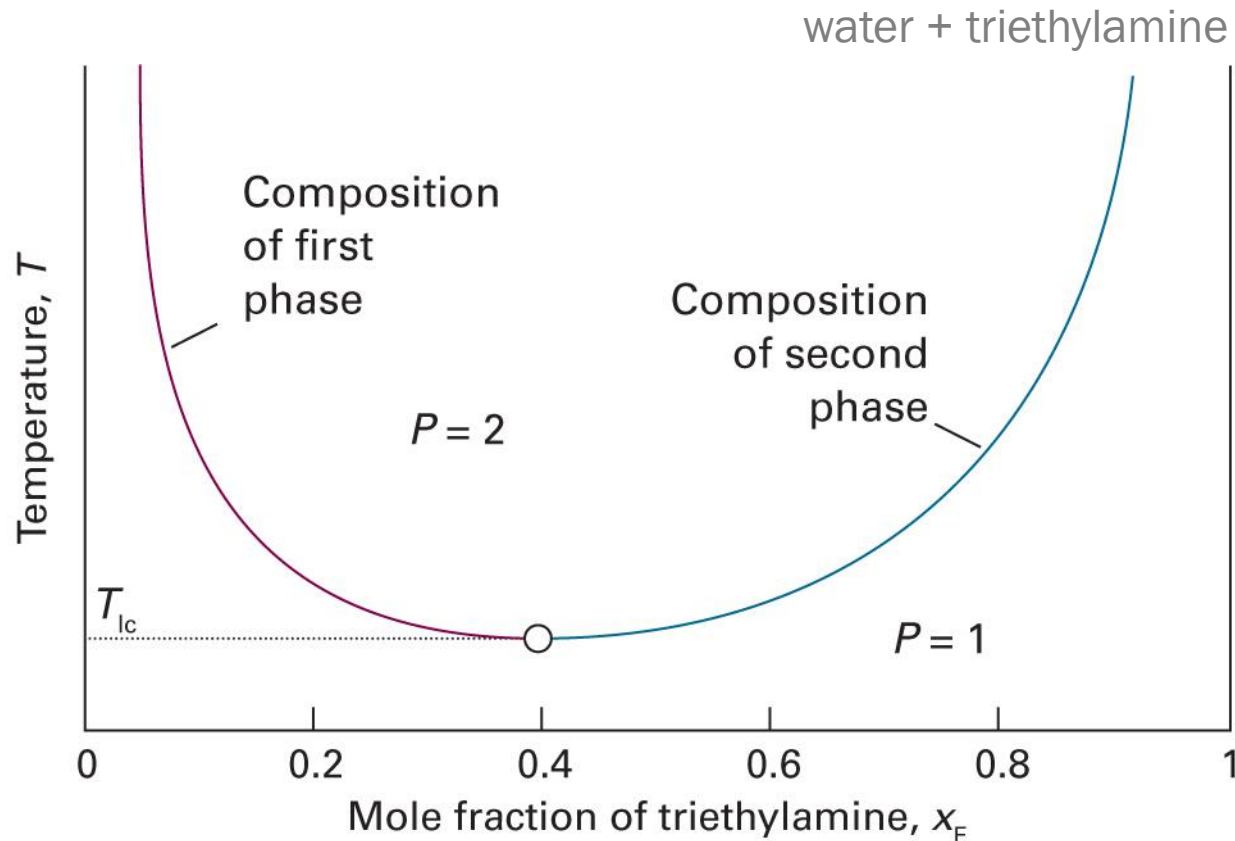
Upper Critical Solution Temperature



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 5C.14)

Last class

Lower Critical Solution Temperature



Two components form **weak complexes** at low T

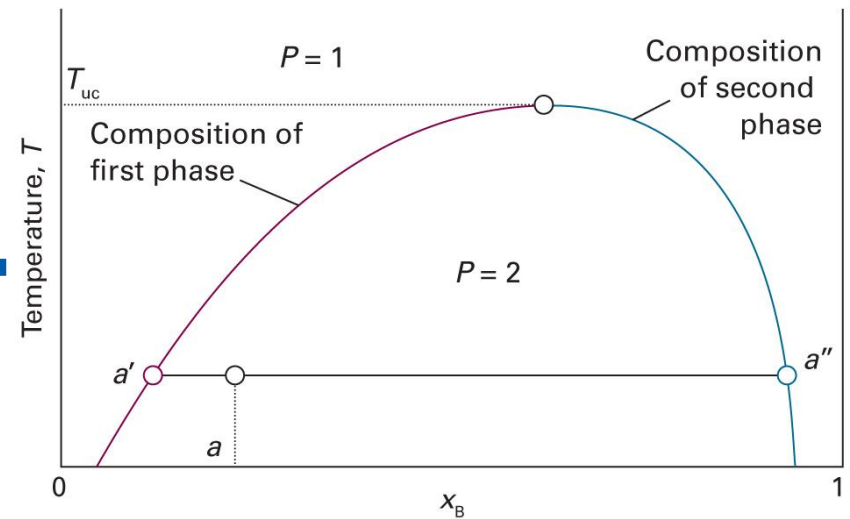
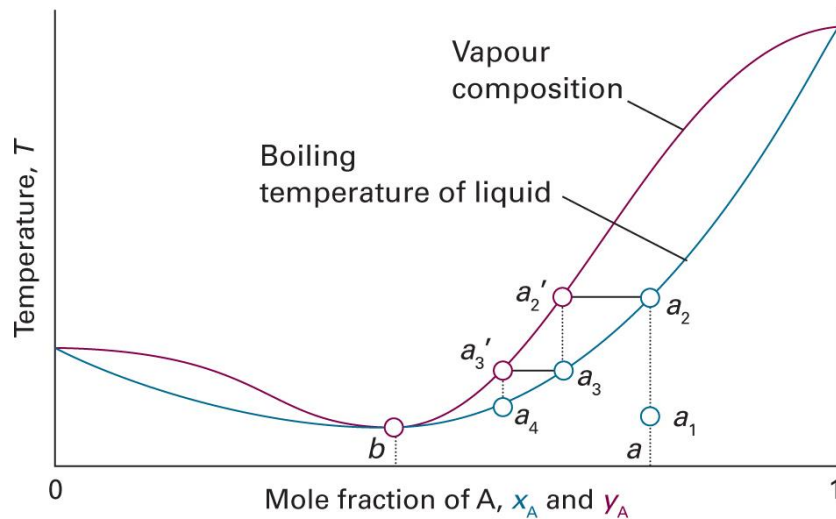
(Image: Atkins *et al.*, *Atkins' Physical Chemistry*, 12th ed., Figure 5C.19)



Last class

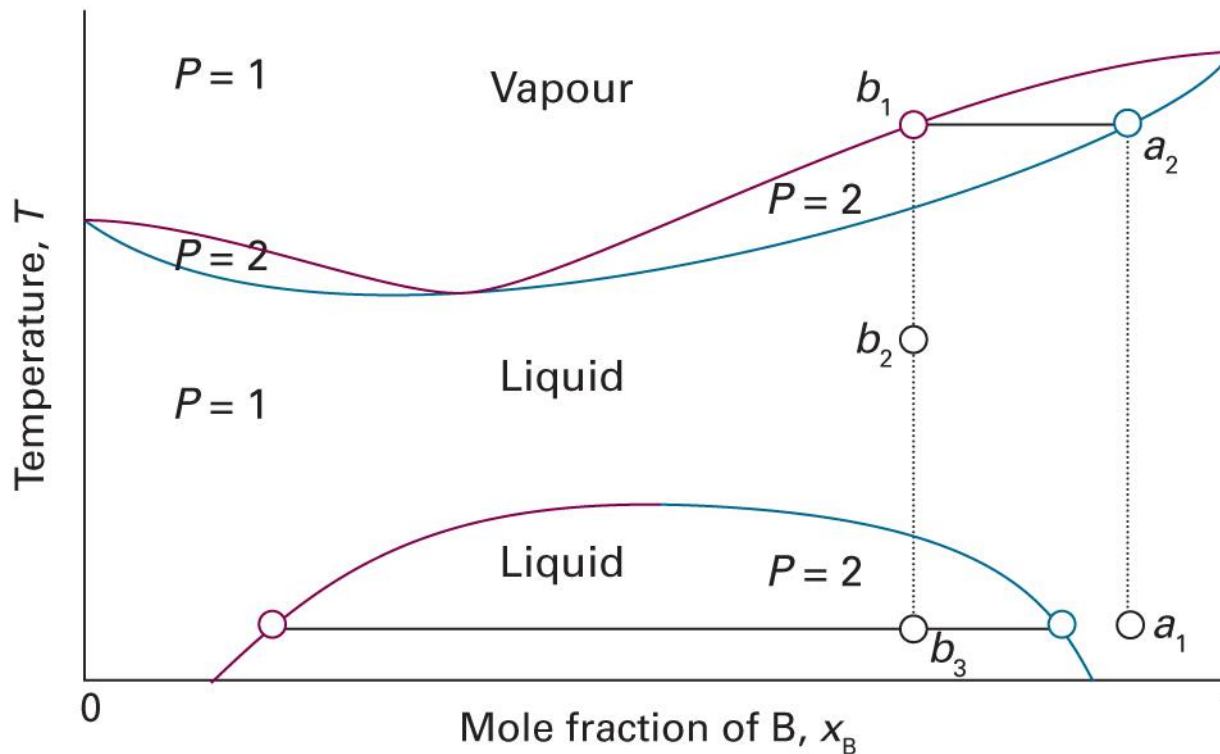
Azeotrope + Phase Separation

(when $\xi \gg 0$)



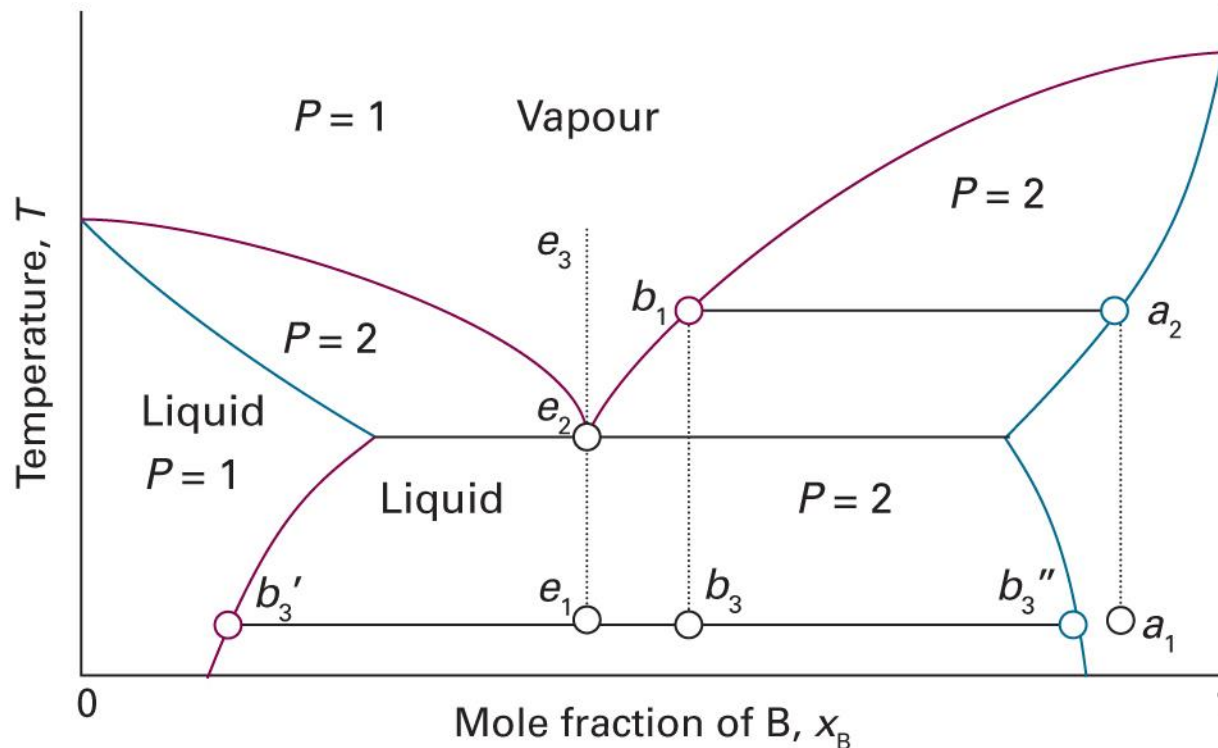
Last class

Case 1: Miscible before Boiling



Last class

Case 2: Boiling before Miscible



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 5C.22)

Liquid-Solid Equilibrium

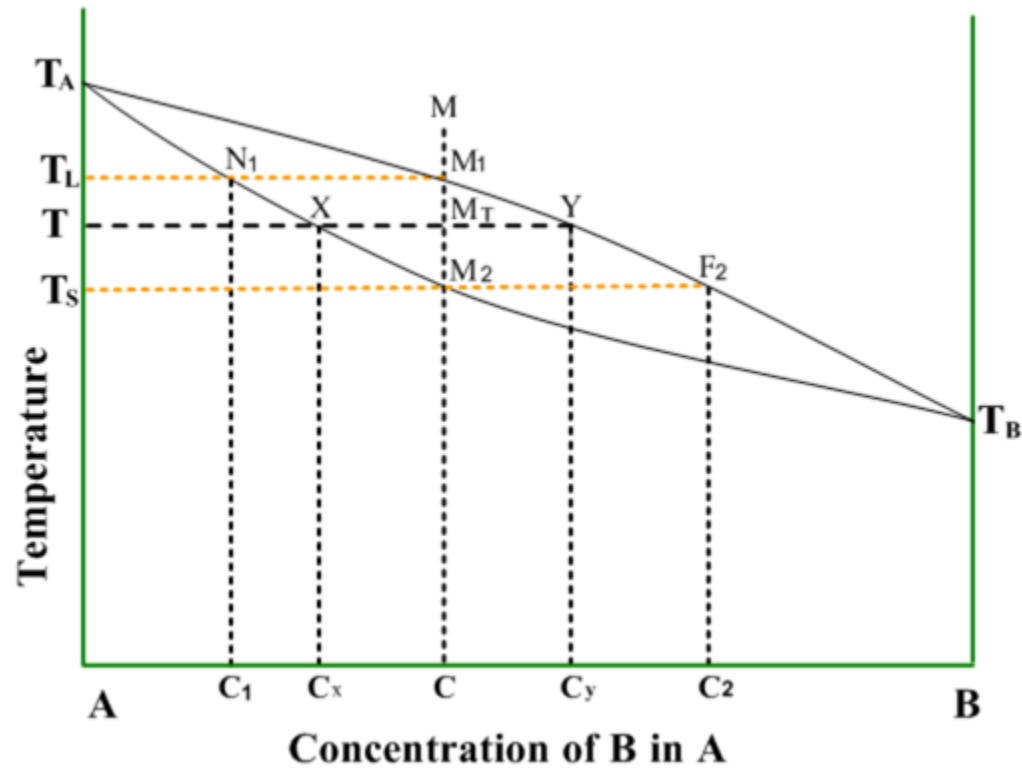
Various scenarios are possible:

1. Miscible liquid + **miscible** solid
2. Miscible liquid + **immiscible** solid
3. Miscible liquid + **completely immiscible** solid

We can also consider **miscibility of liquid**

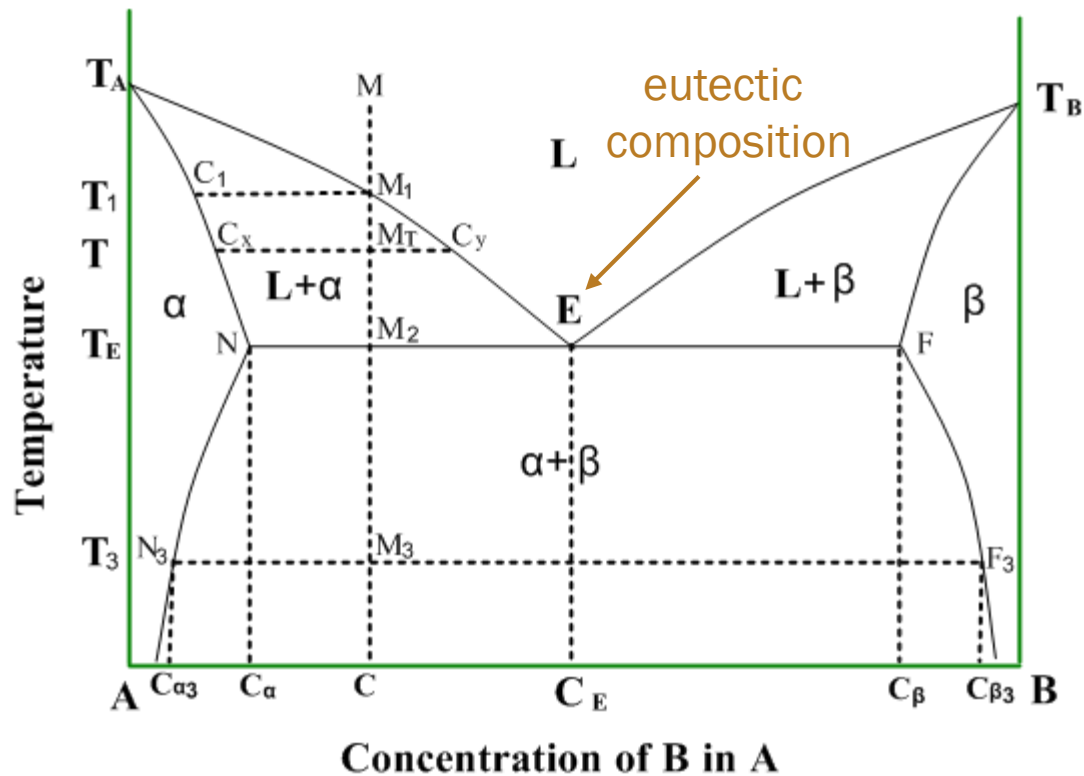
Not in Atkins

Miscible Solid

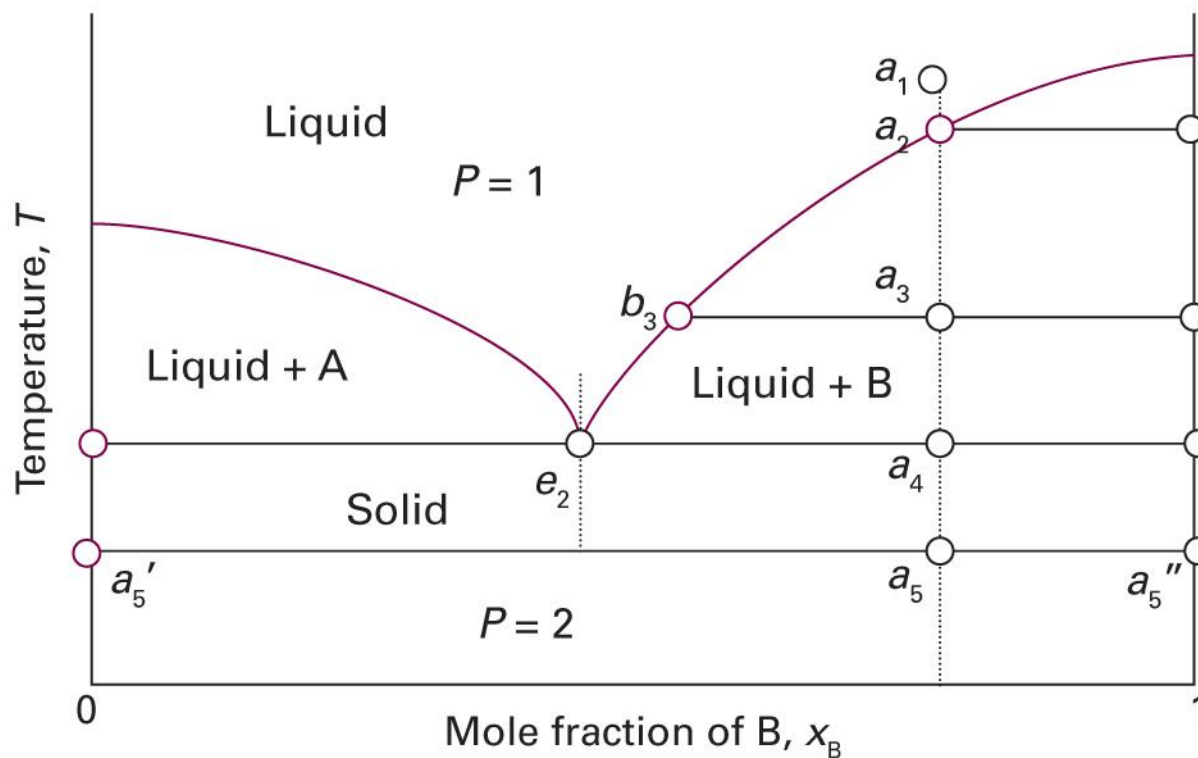


Not in Atkins

Immiscible Solid



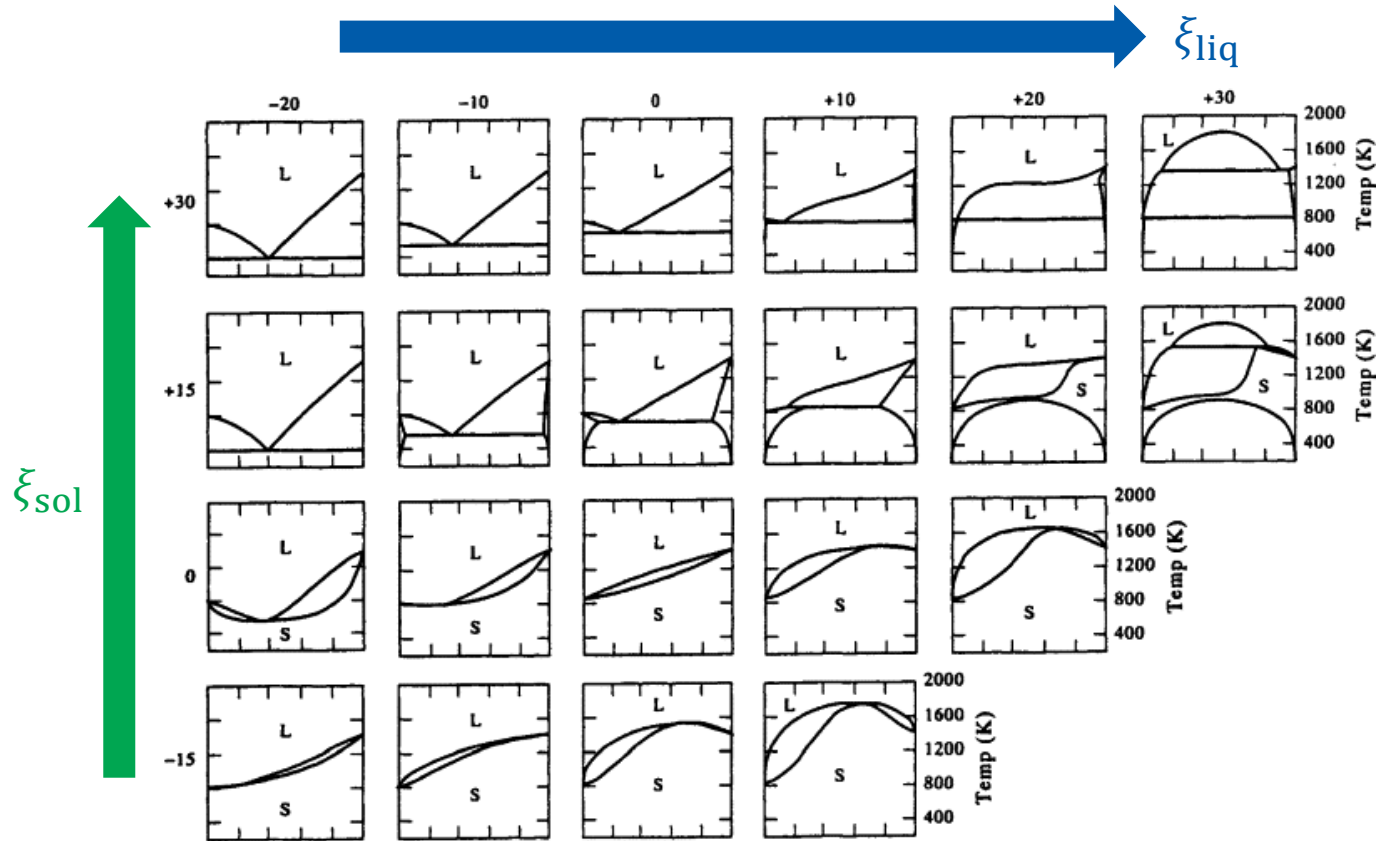
Completely Immiscible Solid



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 5D.1)

Not in Atkins

Diverse Phase Diagrams



(Image: Gaskell, *Introduction to the Thermodynamics of Materials*, 5th Ed., Fig. 10.23)

Reacting Systems

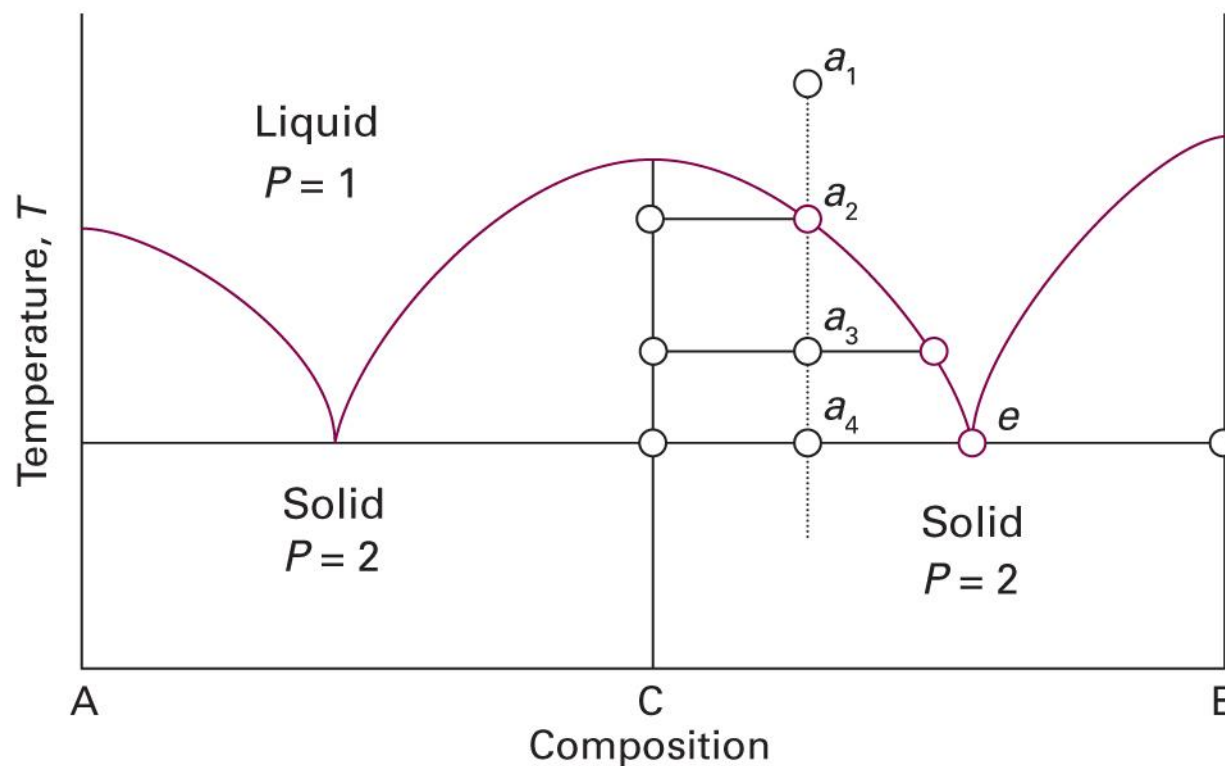
For reaction $A + B \rightarrow C$ (irreversible),

- A 100%, B 0% = A only
- A 75%, B 25% = A $\frac{2}{3}$, C $\frac{1}{3}$
- A 50%, B 50% = C only
- A 25%, B 75% = B $\frac{2}{3}$, C $\frac{1}{3}$
- A 0%, B 100% = B only

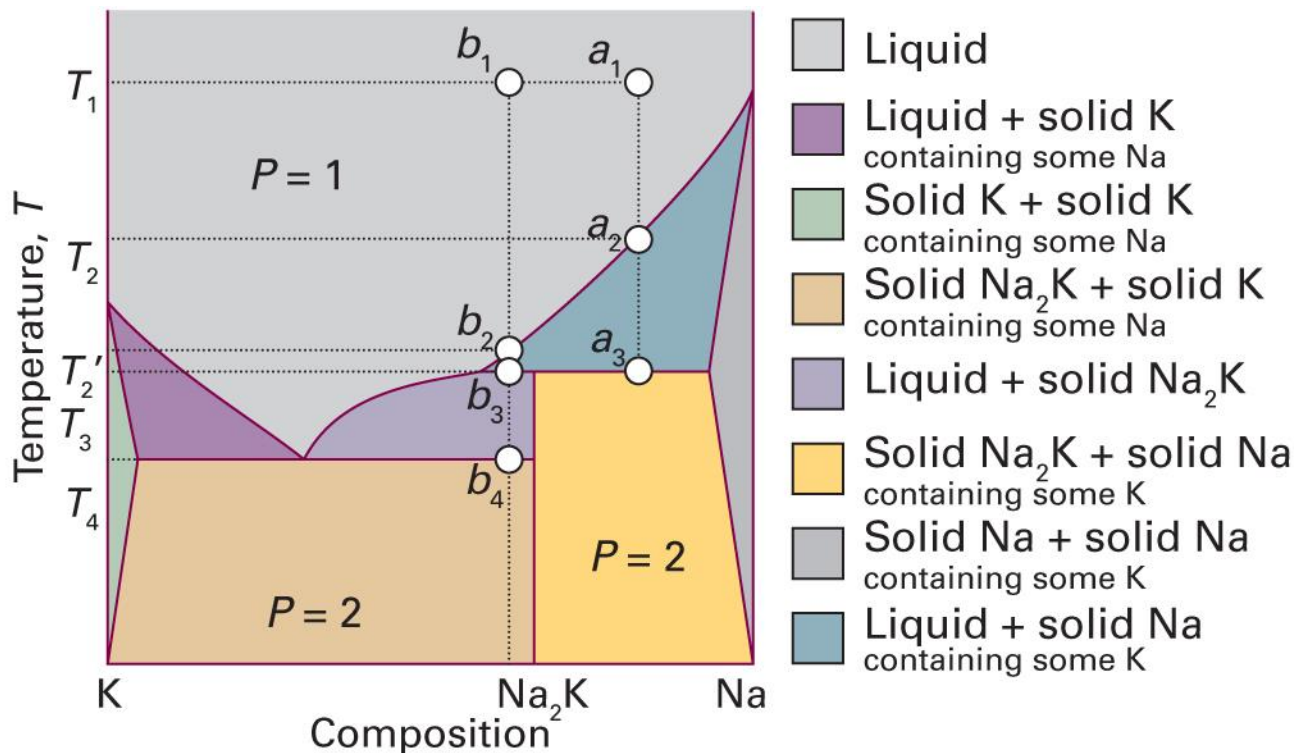
The x-axis can be $(A \rightarrow C \rightarrow B)$

Congruent Melting

(assuming completely immiscible solid)



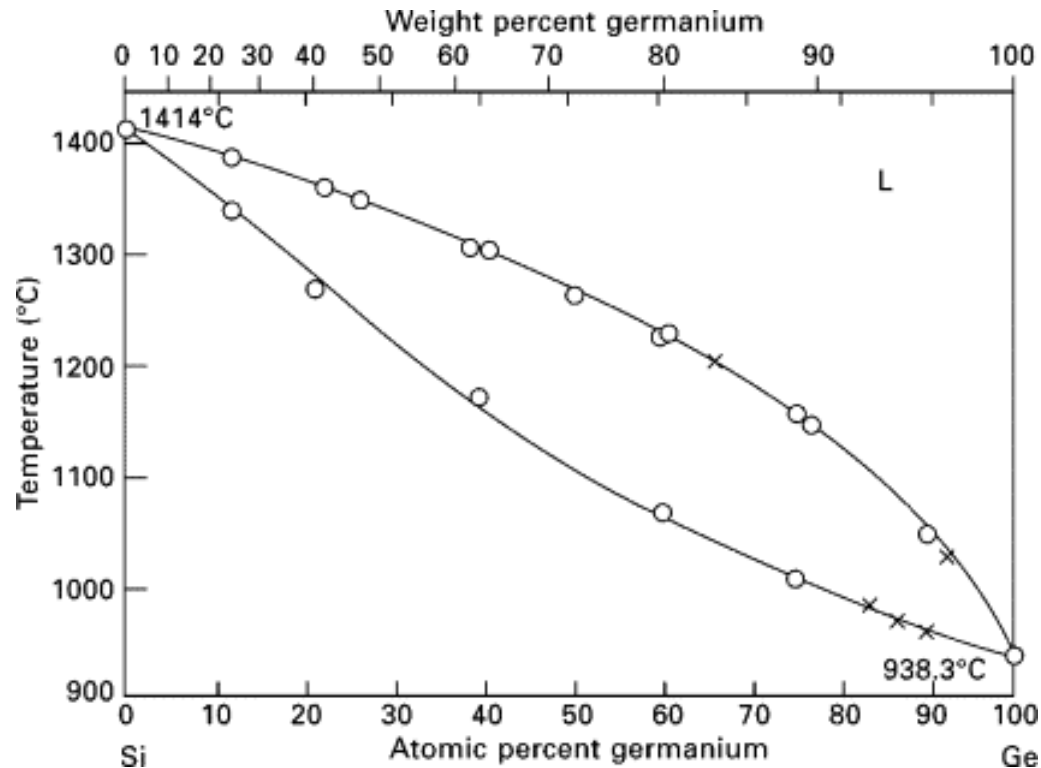
Incongruent Melting



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5D.5)

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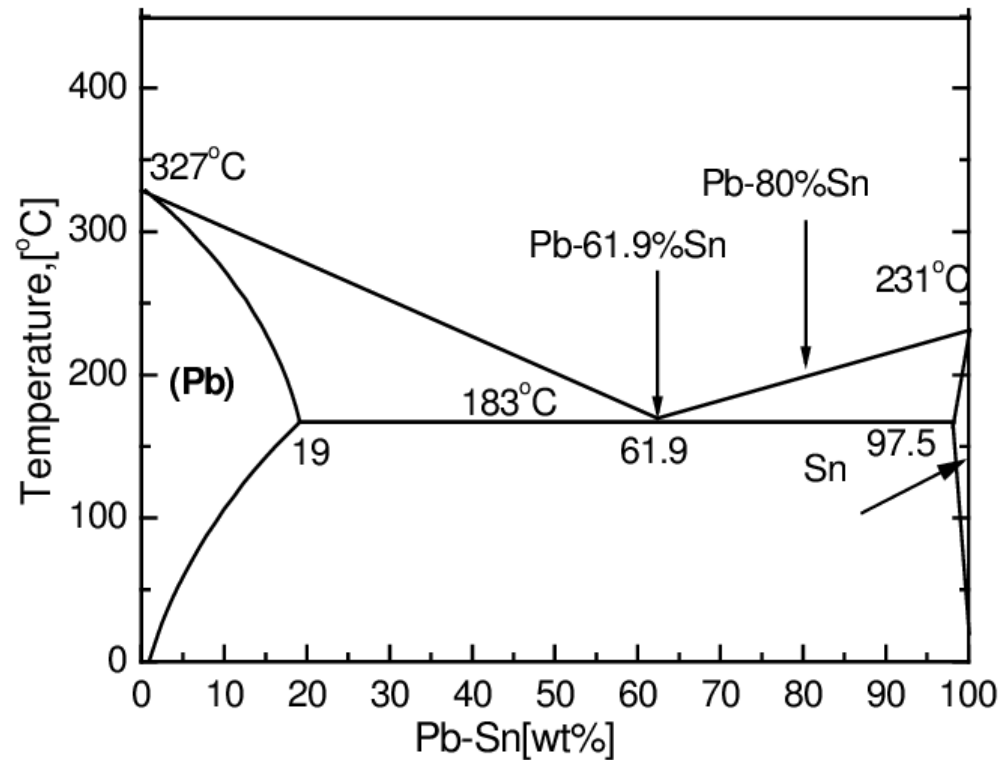
Si-Ge System



(Image: Kasper and Herzog, *Silicon-Germanium (SiGe) Nanostructures* (2011), Fig. 1.2)

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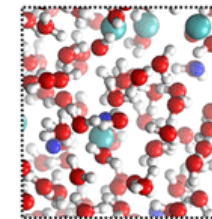
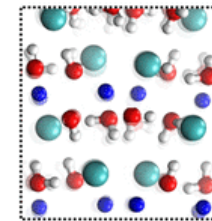
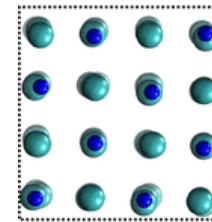
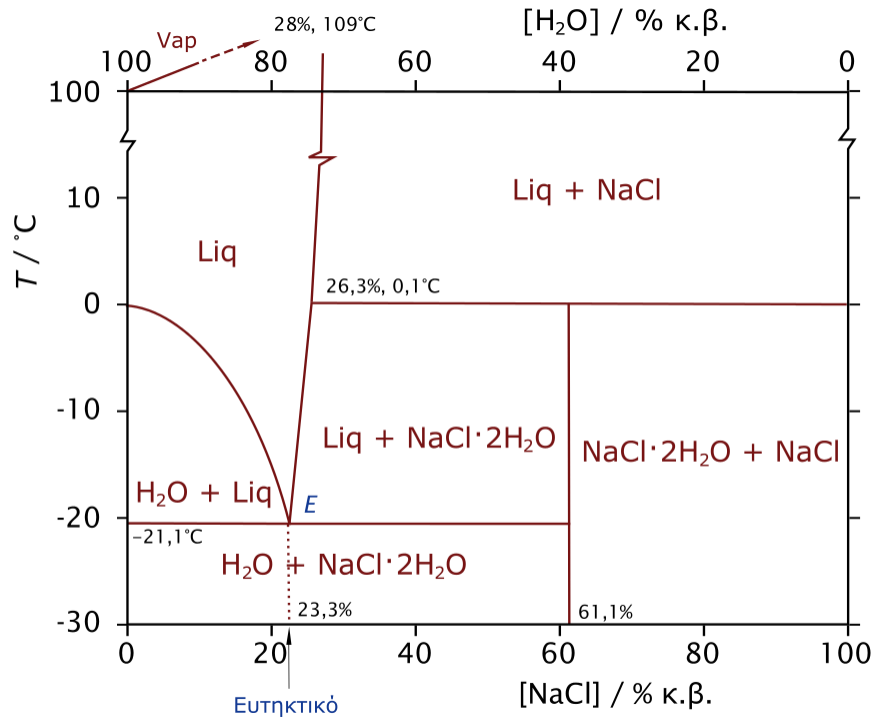
Pb-Sn System



(Image: https://www.researchgate.net/figure/The-Pb-Sn-phase-diagram_fig1_291162603)

Not in Atkins

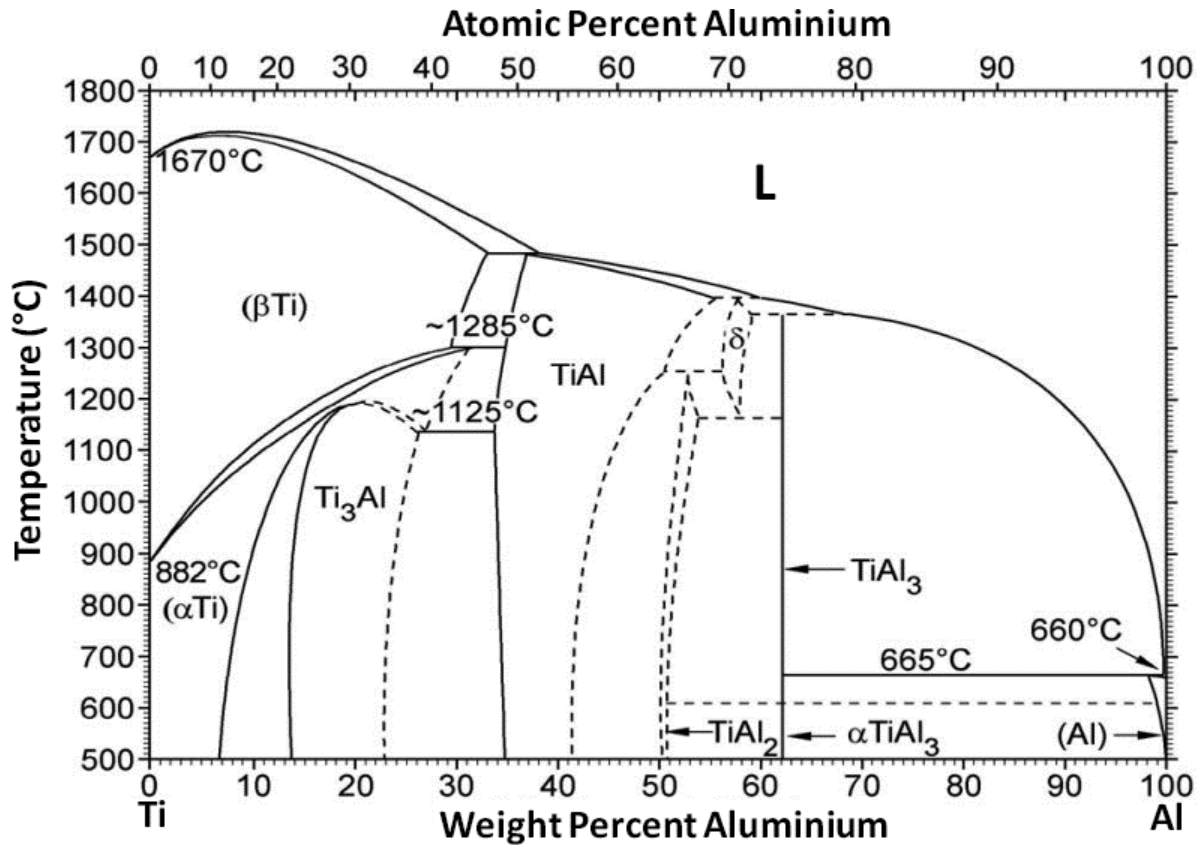
H₂O-NaCl System



(Image: <https://commons.wikimedia.org/wiki/File:H2O-NaCl-phase-diagram-greek.svg>)

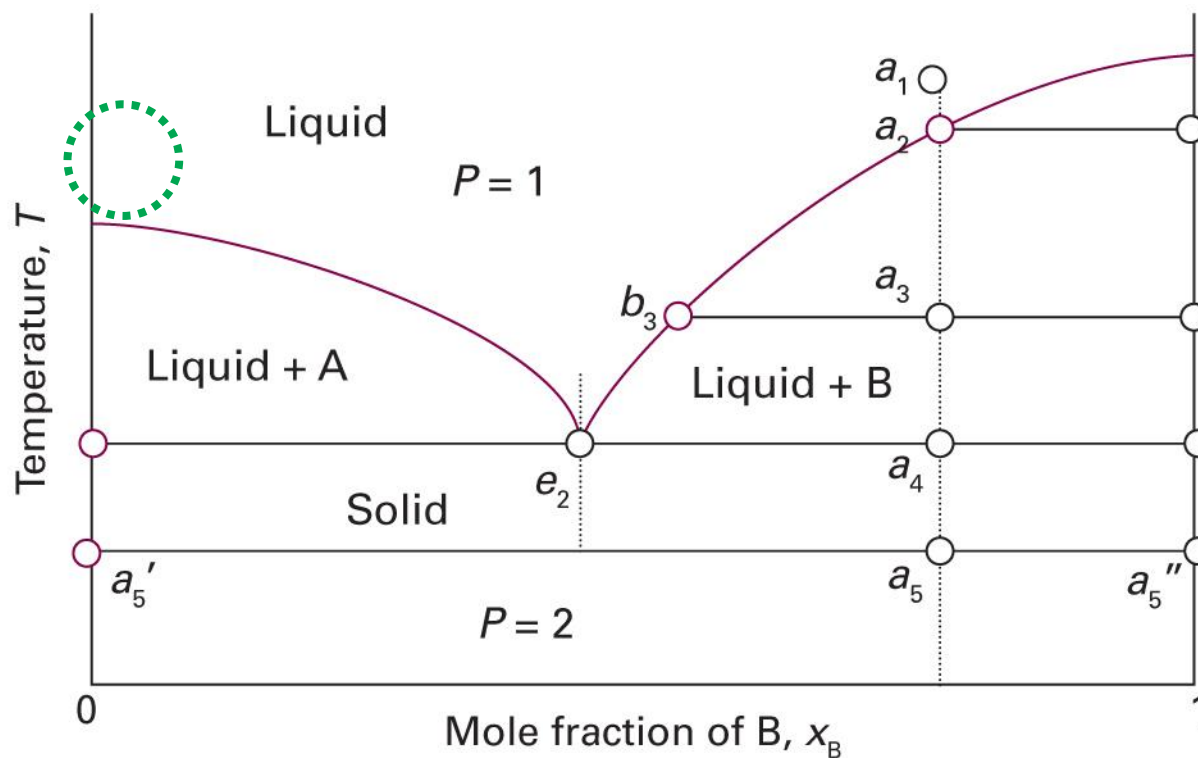
Not in Atkins

Ti-Al System



(Image: Mitra and Wanhill, *Structural Intermetallics* (2016), Figure 10)

A Little Solid in Liquid



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 5D.1)

A Little Solid in Liquid

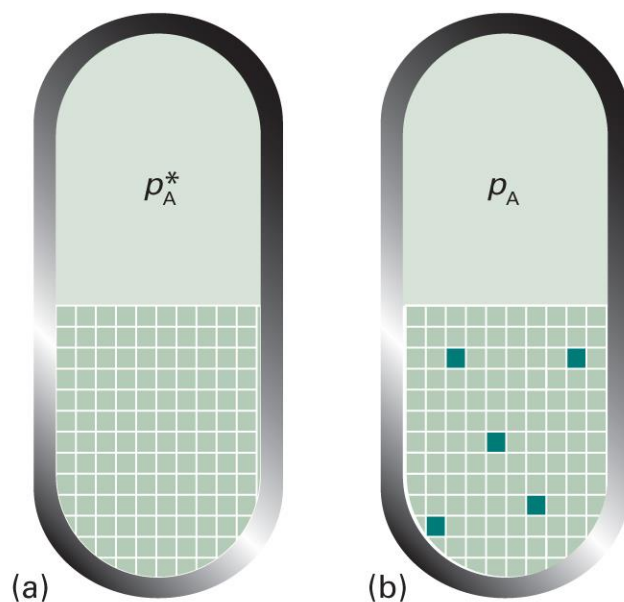
- Considered as an **ideal-dilute solution**
 - Liquid: solvent
 - Solid: solute
- Two assumptions
 - The solute is not volatile
 - The solute does not dissolve in the solid solvent

Colligative Properties

- Physical properties that depends on the relative number of solute particles present but not their chemical identity
1. Lowering of vapor pressure
 2. Elevation of boiling point
 3. Depression of freezing point
 4. Osmotic pressure

Origin of Colligative Properties

- Due to the **reduction of the chemical potential** of the liquid solvent as a result of the presence of solute



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 5B.7)

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Lowering of Vapor Pressure

From Raoult's law,

$$p_A = x_A p_A^*$$

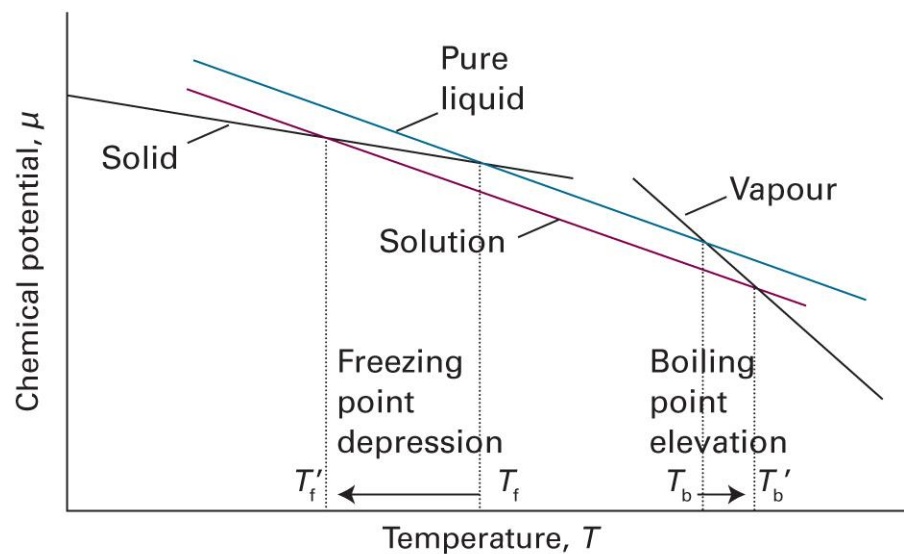
The change in vapor pressure is

$$\Delta p_A = p_A^* - p_A = (1 - x_A) p_A^* = x_B p_A^*$$



Changes in Phase Transitions

$$\mu_A = \mu_A^* + RT \ln x_A \quad (x_A < 1)$$



Boiling: Liquid-Gas Equilibrium

$$\mu_A^*(g) = \mu_A(l)$$

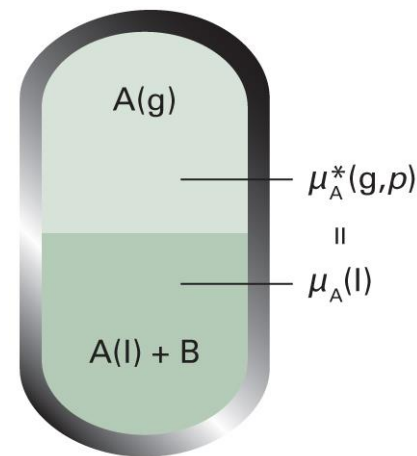
$$\mu_A^*(g) = \mu_A^*(l) + RT \ln x_A$$

which can be rearranged as

$$\ln x_A = \frac{\mu_A^*(g) - \mu_A^*(l)}{RT} = \frac{\Delta_{\text{vap}}G}{RT}$$

Using the Gibbs-Helmholtz equation,

$$\frac{d \ln x_A}{dT} = \frac{1}{R} \frac{d(\Delta_{\text{vap}}G/T)}{dT} = -\frac{\Delta_{\text{vap}}H}{RT^2}$$



Integration

$$\frac{d \ln x_A}{dT} = - \frac{\Delta_{\text{vap}} H}{RT^2}$$

$$d \ln x_A = - \frac{\Delta_{\text{vap}} H}{RT^2} dT$$

$$\int_{\text{pure solvent}}^{\text{current state}} d \ln x_A = - \frac{1}{R} \int_{\text{pure solvent}}^{\text{current state}} \frac{\Delta_{\text{vap}} H}{T^2} dT$$

$$\int_0^{\ln x_A} d \ln x'_A = - \frac{1}{R} \int_{T^*}^T \frac{\Delta_{\text{vap}} H}{(T')^2} dT'$$

Evaluation

$$\int_0^{\ln x_A} d \ln x'_A = -\frac{1}{R} \int_{T^*}^T \frac{\Delta_{\text{vap}} H}{(T')^2} dT'$$

Assuming that $\Delta_{\text{vap}} H$ is constant,

$$\ln x_A = -\frac{\Delta_{\text{vap}} H}{R} \int_{T^*}^T \frac{1}{(T')^2} dT' = -\frac{\Delta_{\text{vap}} H}{R} \left(-\frac{1}{T} + \frac{1}{T^*} \right)$$

$$\ln(1 - x_B) = \frac{\Delta_{\text{vap}} H}{R} \left(\frac{1}{T} - \frac{1}{T^*} \right)$$

Two (Valid) Approximations

$$\ln(1 - x_B) = \frac{\Delta_{\text{vap}}H}{R} \left(\frac{1}{T} - \frac{1}{T^*} \right)$$

For $x_B \ll 1$, $\ln(1 - x_B) \approx -x_B$ and

$$x_B \approx \frac{\Delta_{\text{vap}}H}{R} \left(\frac{1}{T^*} - \frac{1}{T} \right) = \frac{\Delta_{\text{vap}}H}{RTT^*} (T - T^*)$$

Also, supposing $\Delta T_b = T - T^* \ll T^*$,

$$x_B \approx \frac{\Delta_{\text{vap}}H}{R(T^*)^2} \Delta T_b$$

Elevation of Boiling Point

$$x_B \approx \frac{\Delta_{\text{vap}}H}{R(T^*)^2} \Delta T_b$$

Rearrangement gives

$$\Delta T_b = K x_B$$

where $K = R(T^*)^2 / \Delta_{\text{vap}}H$.

Using the molality,

$$\Delta T_b = K_b b$$

where K_b is called the boiling-point constant.



Freezing: Solid-Liquid Equilibrium

$$\mu_A^*(s) = \mu_A(l)$$

$$\mu_A^*(s) = \mu_A^*(l) + RT \ln x_A$$

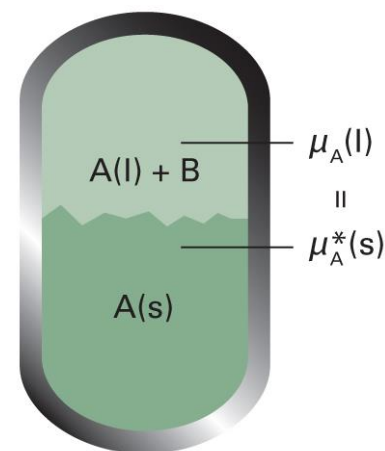
Similarly, we obtain

$$\Delta T_f = K' x_B$$

where $K' = R(T^*)^2 / \Delta_{\text{fus}}H$, or,

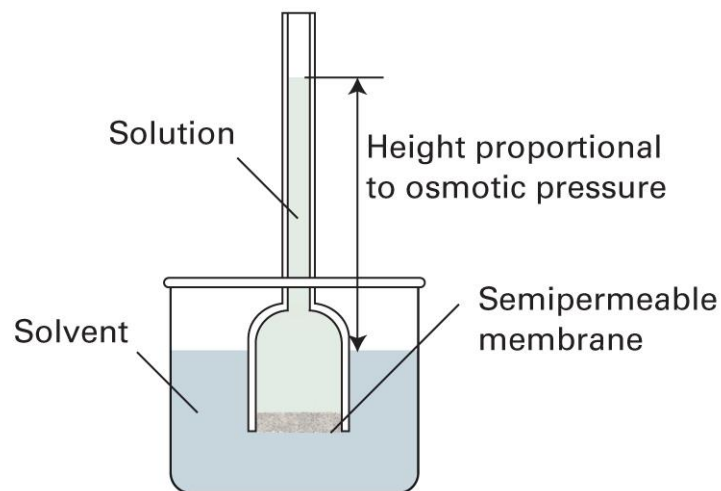
$$\Delta T_f = K_f b$$

where K_f is called the freezing-point constant.

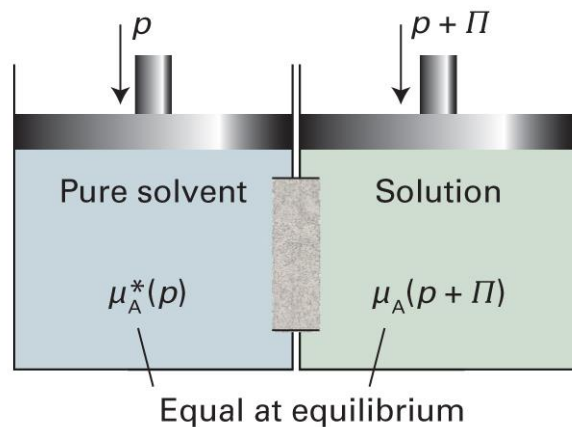


Osmosis

the spontaneous passage of a pure solvent into a solution separated from it by a semipermeable membrane



Thermodynamics of Osmosis



$$\mu_A^*(p) = \mu_A(x_A, p + \Pi)$$

$$\mu_A^*(p) = \mu_A^*(p + \Pi) + RT \ln x_A$$

Thus,

$$\mu_A^*(p + \Pi) = \mu_A^*(p) - RT \ln x_A$$

Pressure Effect

In general,

$$G_m(p_2) = G_m(p_1) + \int_{p_1}^{p_2} V_m dp$$

$$\mu_A^*(p + \Pi) = \mu_A^*(p) + \int_p^{p+\Pi} V_m dp'$$

Assuming that V_m is constant,

$$\mu_A^*(p + \Pi) = \mu_A^*(p) + V_m \int_p^{p+\Pi} 1 dp' = \mu_A^*(p) + V_m \Pi$$

Combining the Two

$$\mu_A^*(p + \Pi) = \mu_A^*(p) - RT \ln x_A$$

$$\mu_A^*(p + \Pi) = \mu_A^*(p) + V_m \Pi$$

Comparing the two equations,

$$-RT \ln x_A = V_m \Pi$$

$$-RT \ln(1 - x_B) = V_m \Pi$$

Since $\ln(1 - x_B) \approx -x_B$ for $x_B \ll 1$,

$$RT x_B = V_m \Pi$$

Osmotic Pressure

$$RTx_B = V_m\Pi$$

For a dilute solution,

$$x_B = n_B/(n_A + n_B) \approx n_B/n_A$$

Thus,

$$RTn_B = n_A V_m \Pi = V \Pi$$

and

$$\Pi = [B]RT \text{ (van't Hoff equation)}$$

Real Solutions

For an ideal solution (now solute = J),

$$\Pi = [J]RT$$

The osmotic virial expansion is

$$\Pi = [J]RT(1 + B[J] + C[J]^2 + \dots)$$

where $B, C, \dots$ are the **osmotic virial coefficients**.

Second Coefficient

$$\Pi = [J]RT(1 + B[J] + \dots)$$

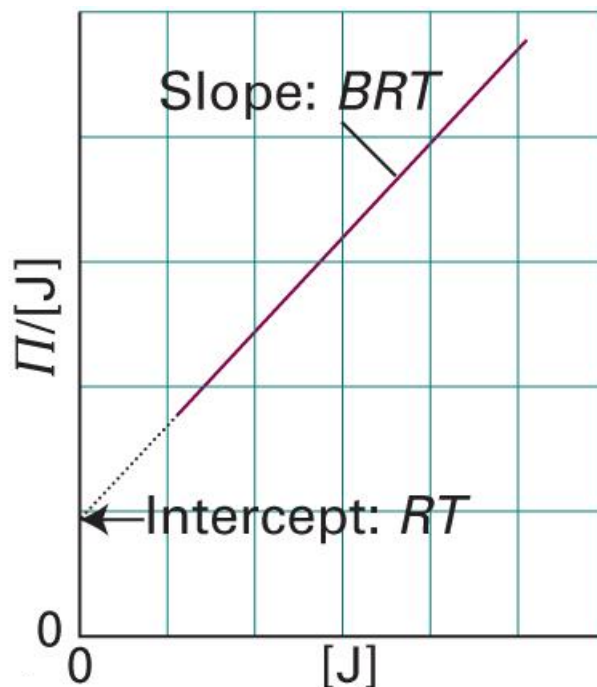
Taking the first correction term only,

$$\Pi \approx [J]RT(1 + B[J])$$

- $B > 0$: solute-solute repulsion dominant
- $B \approx 0$: similar to ideal-dilute solution
- $B < 0$: solute-solute attraction dominant

Osmometry

$$\Pi/[J] \approx RT + BRT[J]$$



Case of Ionic Solutes



- van't Hoff factor i

$$\Delta T_b = iK_b b$$

$$\Delta T_f = iK_f b$$

$$\Pi = i[B]RT$$

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van't Hoff Factors

0.05000 M, 25°C

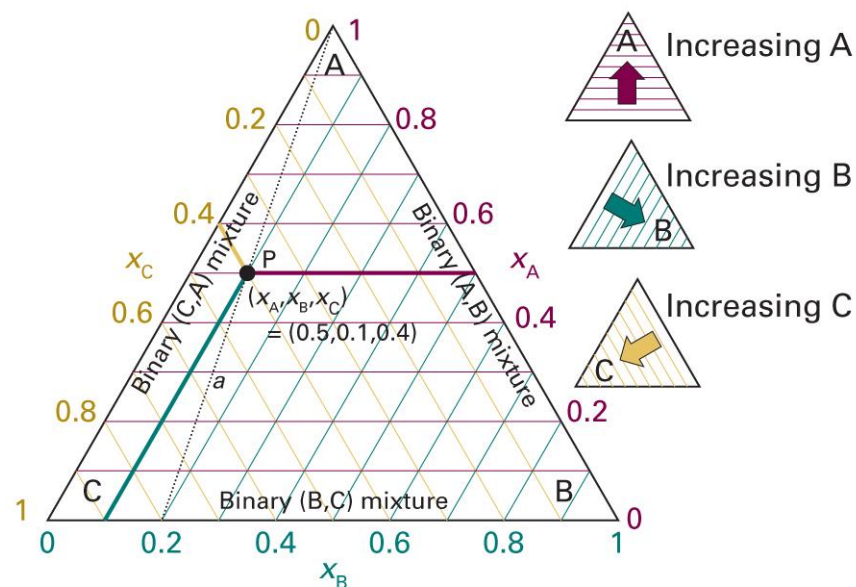
Compound	i (ideal)	i (measured)
glucose	1	1.0
NaCl	2	1.9
KCl	2	1.9
MgCl ₂	3	2.7
FeCl ₃	4	3.4
Ca(NO ₃) ₂	3	2.5
AlCl ₃	4	3.2
MgSO ₄	2	1.4



(Image: <https://chem.libretexts.org/@go/page/24261>)

Ternary Systems: Triangle

$$x_A + x_B + x_C = 1$$

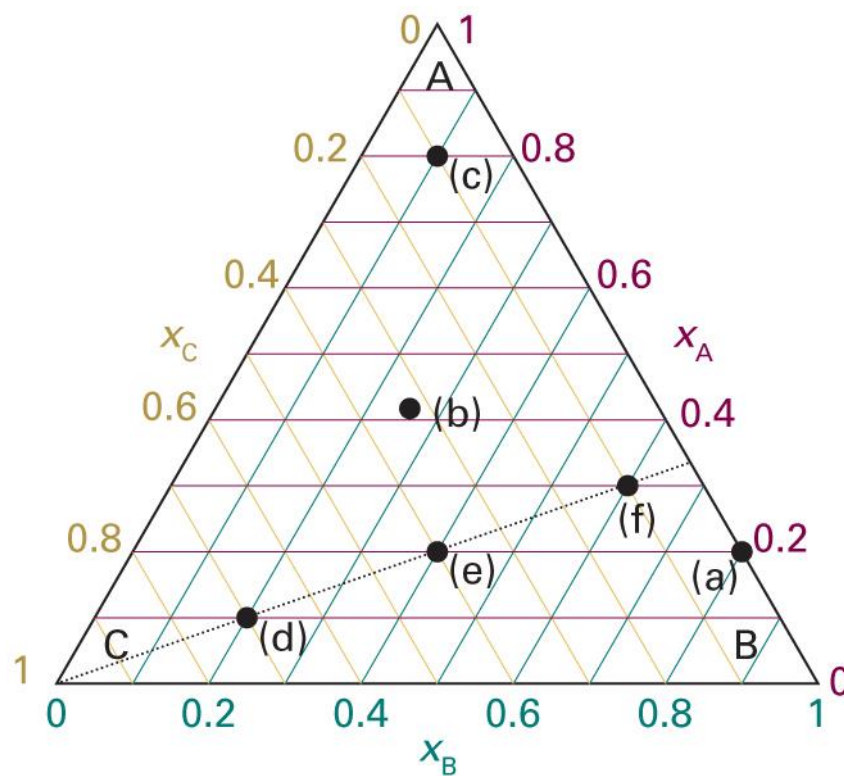


(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 5E.1)

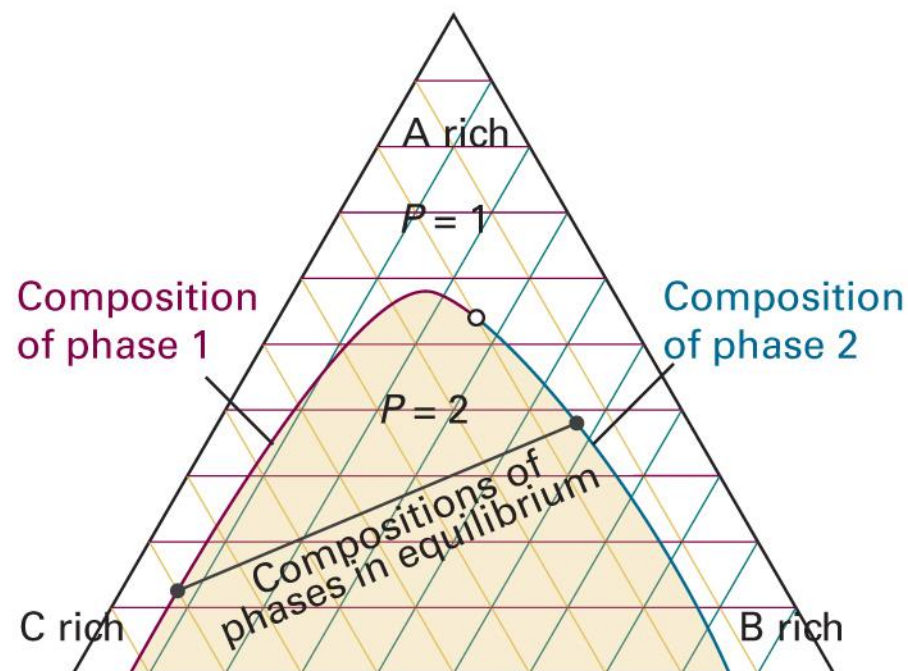
Brief Illustration 5E.1

Indicate the following points:

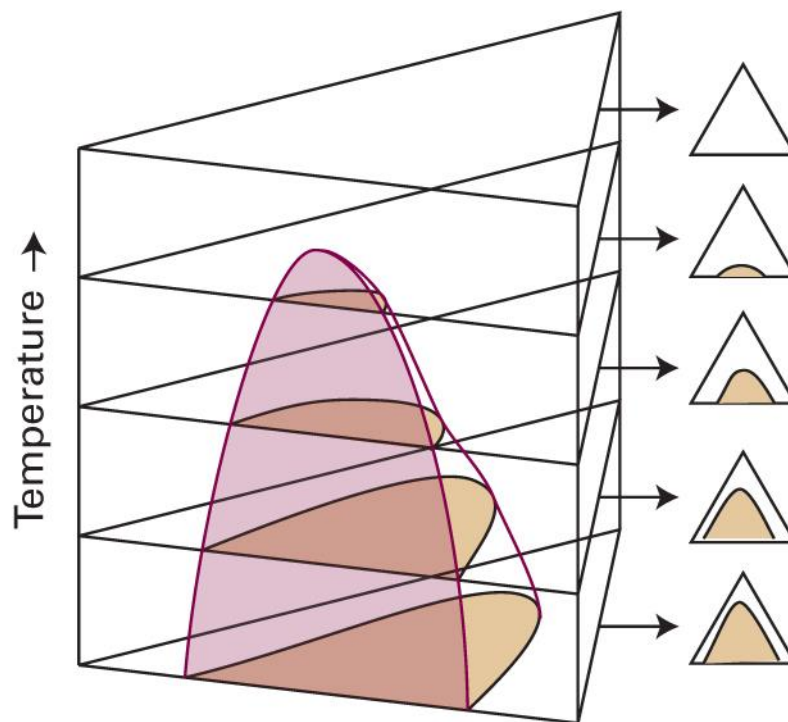
Point	x_A	x_B	x_C
a	0.20	0.80	0
b	0.42	0.26	0.32
c	0.80	0.10	0.10
d	0.10	0.20	0.70
e	0.20	0.40	0.40
f	0.30	0.60	0.10



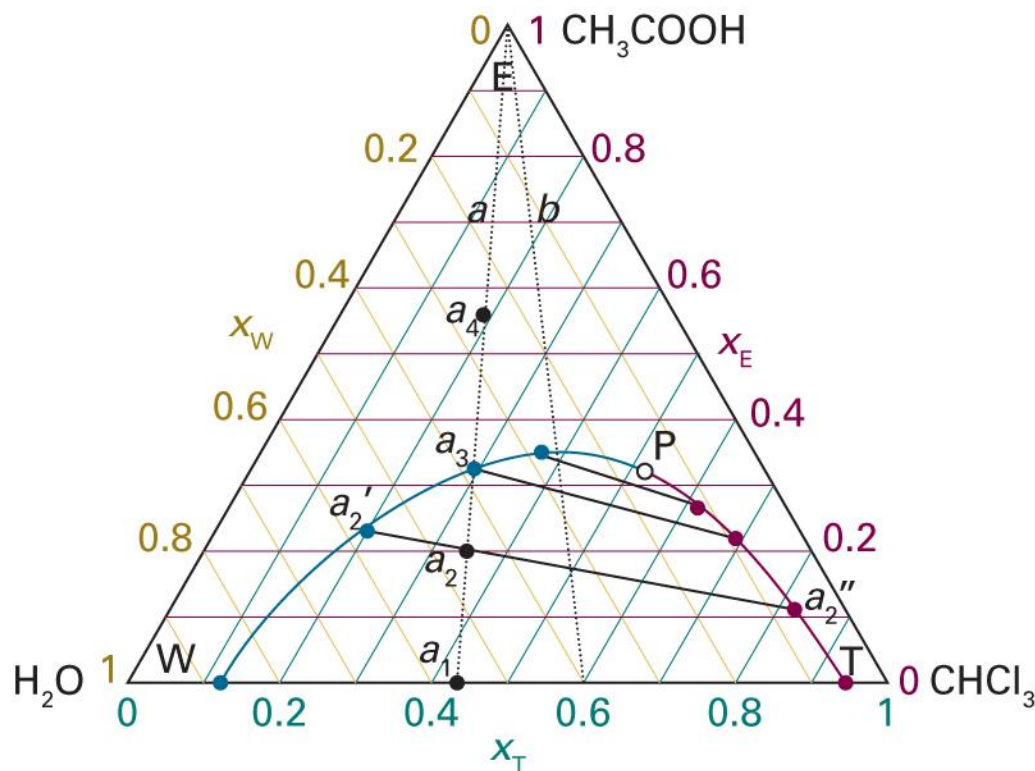
Partially Miscible Liquids



What If Temperature Changes?

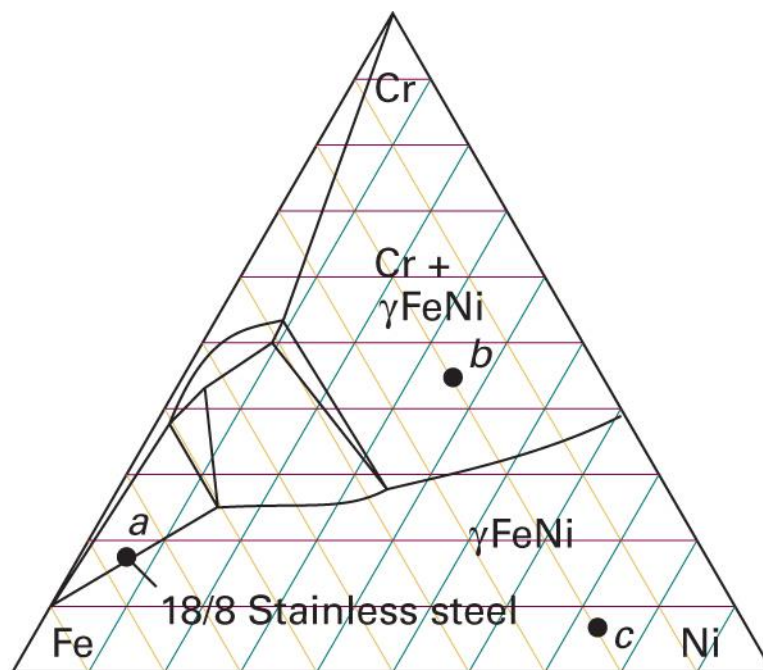


Example: $\text{CH}_3\text{COOH}-\text{H}_2\text{O}-\text{CHCl}_3$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 5E.4)

Ternary Solids



Summary

- Phase
- Chemical potential
- Gibbs-Duhem eq.



Multi-component system

Gas mixture

- Dalton's law
- Perfect gas
- Real gas

Ternary systems

Liquid mixture

Ideal solution

- Raoult's law
$$p_A = x_A p_A^*$$
- Chemical potential
$$\mu_A = \mu_A^* + RT \ln x_A$$
- $\Delta_{\text{mix}}G = -T\Delta_{\text{mix}}S$

Ideal-dilute solution

- Henry's law
$$p_B = x_B K_B$$
- Chemical potential
$$\mu_B = \mu_B^\oplus + RT \ln x_B$$

Real solution

- Regular solution
- Bragg-Williams model
- Excess function
- Activity

$$\mu_A = \mu_A^* + RT \ln a_A$$
$$\mu_B = \mu_B^\oplus + RT \ln a_B$$

- L-G phase diagram
- L-L phase diagram
- L-S phase diagram

Colligative properties

